

CAMBRIDGE DORCHESTER, MD, USA

WMO: 722218

Lat: 38.539N

Lon: 76.030W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 00356

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	20.6	0.3	5.6	21.0	4.9	7.1	24.1	24.6	31.6	22.3	31.5	7.4	310	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	91.2	76.7	90.1	75.8	87.8	74.9	79.7	87.8	78.1	85.6	76.8	84.1	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.2	83.2	76.5	138.3	82.5	74.6	129.4	81.0	43.1	87.2	41.5	86.1	40.1	85.1	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.5	15.6	9.8	95.4	4.6	2.5	6.5	97.2	3.8	98.7	1.2	100.1	-2.1	101.9		
			10.2	81.1	5.2	1.7	6.4	82.3	3.4	83.4	0.4	84.3	-3.4	85.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	35.7	38.6	45.2	56.1	65.4	73.2	77.7	75.5	69.6	59.7	48.1	41.0
	DBStd	16.35	9.08	9.12	8.83	8.46	7.29	5.64	4.80	4.59	5.66	8.18	7.70	8.43
	HDD50	1470	453	336	207	35	0	0	0	0	0	11	125	303
	HDD65	4124	909	738	616	288	90	5	0	0	20	206	508	744
	CDD50	4112	9	17	57	217	476	697	857	792	587	311	68	24
	CDD65	1294	0	0	1	20	102	252	392	327	157	42	1	0
	CDH74	11255	0	0	7	183	731	2219	3880	2847	1131	256	1	0
	CDH80	3502	0	0	0	35	138	682	1461	876	264	46	0	0
Wind	WSAvg	6.0	7.1	7.1	7.8	7.4	5.8	5.0	4.3	4.1	4.8	5.8	6.3	6.2
Precipitation	PrecAvg	46.6	3.6	3.2	4.1	3.5	4.4	3.6	4.9	4.7	3.8	3.3	3.5	3.6
	PrecMax	66.9	7.6	7.8	8.0	7.1	9.8	11.7	12.4	10.7	10.4	8.0	8.5	8.9
	PrecMin	31.7	0.4	0.6	1.8	0.6	0.8	0.6	1.4	0.5	0.6	0.0	0.5	0.6
	PrecStd	8.7	1.5	1.8	1.7	1.5	2.4	2.4	2.5	3.1	2.5	1.9	2.1	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.5	68.2	74.6	83.9	88.0	94.9	97.1	91.5	91.0	84.2	71.9	67.8
		MCWB	58.4	59.8	62.6	68.1	73.5	76.4	80.2	77.1	74.8	72.6	62.0	62.7
	2%	DB	59.2	62.9	69.7	79.1	83.8	90.5	92.5	90.3	86.4	80.7	68.0	62.9
		MCWB	55.2	56.4	59.4	64.5	71.6	75.0	78.3	76.4	73.5	70.3	62.6	60.2
	5%	DB	54.7	57.4	63.9	74.7	81.0	87.8	90.4	87.9	82.3	76.9	63.4	59.1
		MCWB	50.6	51.7	56.0	63.7	69.0	73.8	76.8	75.4	70.9	67.8	58.2	55.4
	10%	DB	50.2	53.6	60.6	71.7	78.7	83.8	88.0	84.4	80.7	72.7	61.0	54.7
		MCWB	46.3	48.1	53.6	61.6	67.3	72.4	75.7	73.8	69.9	65.5	55.8	51.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	62.3	64.7	71.0	75.3	80.3	82.2	81.0	77.4	75.6	66.2	63.9
		MCDB	62.7	65.5	70.8	78.5	83.1	89.4	91.4	88.4	86.1	81.0	69.6	65.0
	2%	WB	56.4	57.9	60.6	67.3	73.2	77.3	80.5	78.8	75.7	72.7	62.9	60.2
		MCDB	59.8	61.8	66.1	75.8	80.8	85.2	89.3	85.4	83.3	77.7	65.4	62.2
	5%	WB	51.8	53.1	57.7	65.0	71.3	75.5	78.7	77.5	74.1	70.1	59.9	56.7
		MCDB	53.9	56.8	63.4	73.4	78.3	84.4	86.9	84.1	80.2	74.5	63.0	59.4
	10%	WB	46.1	48.4	54.7	62.3	69.7	74.2	77.4	76.0	72.4	67.5	56.9	52.2
		MCDB	48.9	52.8	59.0	69.3	75.7	81.9	84.9	82.7	77.9	73.2	60.9	54.0
Mean Daily Temperature Range	5% DB	MDBR	14.1	15.3	16.4	19.0	17.7	17.3	16.9	16.9	17.6	17.6	16.3	14.5
		MCDBR	20.9	22.3	22.8	23.4	19.3	19.6	18.9	18.4	20.4	20.1	19.4	18.5
		MCWBR	18.0	16.9	15.0	12.4	9.5	8.0	7.9	7.7	9.6	11.4	14.0	16.6
	5% WB	MCDBR	19.5	21.1	21.0	21.0	16.9	17.4	17.2	15.9	17.2	16.4	15.7	17.1
		MCWBR	18.5	17.8	15.6	13.4	9.4	8.1	8.0	7.5	9.3	10.8	13.6	16.8
Clear-Sky Solar Irradiance	taub	0.313	0.320	0.356	0.396	0.441	0.488	0.516	0.508	0.413	0.360	0.327	0.312	
	taud	2.484	2.459	2.405	2.312	2.218	2.132	2.069	2.077	2.320	2.469	2.520	2.527	
	Ebn at Noon	275	288	287	280	268	254	246	244	264	270	268	267	
	Edh at Noon	25	30	34	40	45	49	52	50	37	29	24	23	
All-Sky Solar Radiation	RadAvg	689	954	1273	1614	1832	2013	1957	1749	1421	1072	777	606	
	RadStd	55	100	107	116	207	152	94	103	147	114	68	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (86 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+151	

Nomenclature: See separate page